

BILLINGS LOGAN, MT, USA

WMO: 726770

Lat: 45.807N

Lon: 108.542W

Elev: 3581

StdP: 12.89

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24033

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.6	-2.7	-14.0	2.9	-7.4	-8.7	3.9	-1.6	33.0	30.7	29.5	32.7	7.7	30	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.6	94.9	62.6	91.7	61.9	88.4	61.3	66.3	85.1	64.7	84.0	63.2	82.9	9.1	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.4	89.5	72.2	58.2	82.6	70.8	56.2	77.0	69.9	32.9	84.7	31.5	83.5	30.4	82.8	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	25.0	22.3	-15.1	100.6	6.1	2.7	-19.5	102.5	-23.1	104.1	-26.6	105.6	-31.0	107.6
			-15.3	69.6	5.7	2.5	-19.3	71.4	-22.6	72.9	-25.8	74.3	-29.9	76.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	27.2	28.9	37.8	45.9	55.1	64.7	74.1	71.9	61.7	48.3	36.6	27.3
	DBStd	19.47	13.32	13.94	12.59	9.47	8.34	7.34	5.93	6.32	9.11	9.86	12.23	12.92
	HDD50	3210	708	593	398	183	44	3	0	0	16	146	416	703
	HDD65	6746	1173	1010	845	575	321	91	6	18	164	523	852	1168
	CDD50	2628	1	3	18	59	202	444	747	680	367	92	15	0
	CDD65	687	0	0	0	1	14	82	288	233	65	4	0	0
	CDH74	8225	0	0	3	36	223	982	3385	2701	839	56	0	0
	CDH80	3618	0	0	0	3	48	356	1657	1234	310	10	0	0
Wind	WSAvg	10.6	13.4	12.0	10.8	10.5	9.9	9.4	8.8	8.8	9.2	10.5	11.9	12.7
Precipitation	PrecAvg	14.3	0.7	0.6	1.0	1.7	2.5	2.0	1.0	0.9	1.3	1.2	0.7	0.7
	PrecMax	26.8	2.4	2.0	2.7	4.4	9.5	6.3	5.1	3.5	4.3	3.8	2.3	2.0
	PrecMin	7.1	0.0	0.0	0.1	0.0	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.1
	PrecStd	3.5	0.5	0.4	0.6	1.0	1.8	1.5	0.9	0.8	1.0	0.8	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	61.8	70.9	78.8	85.4	94.6	99.9	98.0	92.7	80.6	67.7	55.7
		MCWB	41.3	44.0	48.4	53.2	57.6	62.4	64.0	63.2	60.7	56.2	48.0	41.9
	2%	DB	50.1	55.2	65.4	72.4	79.9	88.6	95.9	93.9	88.0	73.6	61.1	50.2
		MCWB	38.8	41.1	46.2	51.1	57.0	62.6	63.4	62.0	59.3	52.5	44.9	38.9
	5%	DB	46.3	50.3	59.8	67.2	75.5	83.7	92.7	90.6	83.2	68.2	56.7	46.7
		MCWB	36.8	38.6	44.0	48.7	55.8	61.1	62.5	61.2	57.6	50.3	43.1	36.8
	10%	DB	43.2	46.4	54.9	62.0	71.0	79.6	89.3	87.3	78.6	63.0	52.5	43.2
		MCWB	34.8	36.5	41.5	46.6	53.9	59.8	62.0	60.7	56.4	48.2	40.8	34.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.3	45.4	49.7	55.2	62.1	67.9	69.2	67.5	65.2	57.3	49.2	43.0
		MCDB	53.1	60.6	67.7	74.2	78.0	82.8	88.3	86.9	82.2	76.8	64.7	53.9
	2%	WB	39.4	41.7	47.0	52.1	59.1	64.8	66.9	65.1	61.8	54.0	46.1	39.6
		MCDB	48.5	53.5	63.1	69.0	74.1	81.9	86.2	85.6	80.3	69.6	59.4	49.0
	5%	WB	37.3	39.2	44.6	49.8	57.0	62.7	65.3	63.5	59.5	51.3	43.7	37.3
		MCDB	45.5	49.4	58.3	65.3	72.2	79.6	84.6	84.4	79.2	65.5	55.4	45.8
	10%	WB	35.2	36.7	42.3	47.5	54.9	60.9	63.8	62.0	57.4	49.2	41.5	35.0
		MCDB	42.7	46.0	53.9	60.4	68.8	76.8	84.0	82.3	75.5	62.0	51.7	42.5
Mean Daily Temperature Range	5% DB	MDBR	16.6	17.8	20.4	20.7	21.3	23.0	26.6	26.4	23.7	19.9	17.4	15.2
		MCDBR	19.1	22.4	27.4	29.7	28.9	30.2	31.1	31.2	30.9	27.4	22.6	18.3
		MCWBR	11.9	12.8	14.0	14.0	12.5	11.5	9.9	10.7	12.1	13.1	12.2	11.1
	5% WB	MCDBR	18.9	21.6	26.0	27.0	26.1	26.7	27.7	27.5	27.7	25.0	21.7	17.7
	MCWBR	12.4	12.8	13.8	13.3	12.1	11.4	10.2	10.3	11.2	12.5	12.3	11.1	
Clear-Sky Solar Irradiance	taub		0.248	0.254	0.269	0.299	0.321	0.329	0.344	0.351	0.316	0.284	0.256	0.243
	taud		2.570	2.577	2.587	2.491	2.484	2.500	2.472	2.414	2.512	2.578	2.578	2.564
	Ebn at Noon		284	302	310	306	300	297	291	285	288	285	277	274
	Edh at Noon		19	23	26	32	33	33	34	34	28	23	19	17
All-Sky Solar Radiation	RadAvg		540	848	1198	1509	1781	2130	2252	1891	1439	909	578	419
	RadStd		41	63	93	86	184	152	89	97	99	69	48	28

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page